Commodity Trading

Multivariate Statistical Arbitrage

Market-Neutral Portfolio Optimization with Gurobi

ESILV — Commodity Markets and Models

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Group: Brent Crude

Anchor: BZ=F (Brent Crude Oil Futures)

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Reference: Yang, H. & Malik, S. (2024). *Optimal Trading Technique for Statistical Arbitrage.* https://github.com/Hongshen-Yang/optimal-trading-technique

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1. Executive Summary

This report presents the design, implementation, and results of a multivariate statistical arbitrage strategy applied to the Brent Crude oil sector. Following the methodology of Yang & Malik (2024), i construct a market-neutral basket of 5 stocks cointegrated with the Brent Crude Futures (BZ=F), and optimize portfolio weights using Gurobi's quadratic programming solver under a strict beta-neutrality constraint ($\sum \beta = 0$).

The strategy exploits mean reversion in the cointegrated spread between the anchor and the basket, generating returns that are entirely decorrelated from the oil market direction. All profits represent **pure alpha** — they come exclusively from the statistical relationship between assets, not from market exposure.

Table 1: Final performance comparison — three-way benchmark

Metric	**Optimized ($\sum\beta$=0)**	**Naive (Equal-Wt)**	**Buy & Hold**
Total Return	+24.42%	–4.02%	+220.92%
CAGR	+5.10%	–0.86%	+30.43%
Ann. Volatility	8.06%	5.23%	25.13%
Sharpe Ratio	0.224	–0.809	0.998
Max Drawdown	–10.17%	–12.48%	–26.22%
Calmar Ratio	0.502	–0.069	1.161
Days in Drawdown	1,001	1,169	986
Final Capital	$1,244,209	$959,846	$3,209,175

The optimized basket outperforms the naive benchmark by **+28.44 percentage points**, proving that the Gurobi optimization and $\sum \beta = 0$ constraint add substantial value. While the Buy & Hold achieves higher absolute returns (benefiting from the 2021–2026 oil bull market, particularly amid the recent Iranian conflict in march 2026), my strategy delivers returns with **3× lower volatility** and **2.5× shallower drawdowns**.

2. Data Architecture and Investment Universe

2.1 Three-Tier Data Structure

Following the project guidelines, the dataset is organized into three mandatory tiers to ensure both statistical robustness and proper sector representation:

Table 2: Three-tier dataset structure (23 assets total)

Tier	Role	Assets	Count
Primary Anchor	Core commodity exposure	BZ=F (Brent Crude Futures)	1
Sector Proxies	Broader sector beta	XLE, XOP, OIH, IEZ, AMLP	5
Individual Stocks	Commodity value chain	BP, SHEL, TTE, CNQ, OXY, ...	17
		Total	**23**

2.2 Data Sourcing and Technical Challenges

All data was downloaded via the Yahoo Finance API (`yfinance`) covering a 5-year period (March 2021 – March 2026) with daily OHLCV granularity. Several technical challenges were encountered and resolved:

- **MultiIndex formatting issue:** A recent Yahoo Finance API update changed the column structure to MultiIndex, which initially produced an empty CSV file (722 bytes). This was resolved by rewriting the extraction script using the `stack()` function to flatten the data into the long format required by the backtester.
- **Delisted/unavailable tickers:** Three tickers (ENI, TLW, REP) failed to download due to exchange-specific ticker formats. The universe was expanded with additional Brent-exposed companies (SU, CNQ, GALP.LS, IMB.L, EC, YPF, E, CTRA, DVN, OXY, SLB) to maintain 15+ assets for statistical robustness.
- **Temporal synchronization:** After downloading, the price matrix was cleaned by removing dates with any missing values, resulting in **1,226 clean trading days** across all 23 assets. This ensures perfect alignment for all statistical computations.

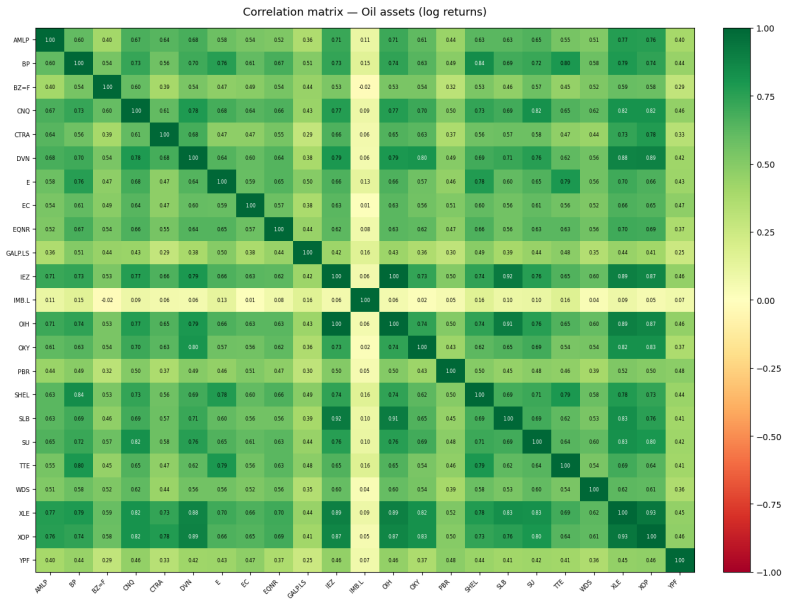

Figure 1: Correlation matrix of log returns across 23 oil sector assets

3. Statistical Analysis and Signal Construction

3.1 Beta Estimation

For each stock in the universe, i compute its rolling beta relative to the Brent anchor (BZ=F) using a 120-day window. Beta measures the sensitivity of each stock's returns to the oil market:

$$\beta_i = \frac{\mathrm{Cov}(r_i,\ r_{\text{anchor}})}{\mathrm{Var}(r_{\text{anchor}})} \tag{1}$$

A stock with $\beta = +0.4$ moves approximately 0.4% for every 1% move in Brent. All stocks in my universe have positive betas (ranging from -0.04 to $+0.44$), meaning they all tend to rise when oil rises. This is precisely the market exposure that the Gurobi optimizer must neutralize.

3.2 Multivariate Cointegration (Johansen Test)

Rather than testing pairwise cointegration, i use the Johansen multivariate cointegration test on groups of 5 stocks plus the anchor. The test determines whether a linear combination of the log-prices forms a stationary (mean-reverting) spread.

We tested **2,000* random combinations** of 5 stocks from the 19-stock universe and found **269 cointegrated baskets**. The selection criteria were:

1. The trace statistic must exceed the 95% critical value.

2. The ADF test on the resulting spread must confirm stationarity.
3. The half-life of mean reversion must be below 60 days.

Table 3: Selected basket characteristics

Metric	**Value**
Selected Basket	CNQ, E, OXY, WDS, YPF
Trace Statistic	116.88 (critical: 95.75)
ADF p-value	0.0000
Half-life	9.2 days
Cointegration Strength	1.22× critical value

The half-life of 9.2 days is particularly favorable — it means the spread reverts to its mean in about two trading weeks, providing frequent trading opportunities.

*:1000 : bad baskets - 2000 : the best compromise - 3000 was useless i got small improvements for 10sec more of calculus

3.3 Spread Construction and Z-Score

The spread is constructed using the first eigenvector from the Johansen decomposition. This cointegrating vector weights each asset's log-price to produce a stationary time series:

$$\text{spread}_t = \sum_i v_i \cdot \log(P_{i,t}) \quad \text{where } \mathbf{v} \text{ is the cointegrating vector} \tag{2}$$

A rolling Z-score (30-day window) normalizes the spread for signal generation:

$$Z_t = \frac{\text{spread}_t - \mu_{30d}}{\sigma_{30d}} \tag{3}$$

Trading signals are generated when the Z-score exceeds the entry threshold ($\pm 1.75\sigma$), and positions are closed when the spread reverts to the mean ($Z = 0$).

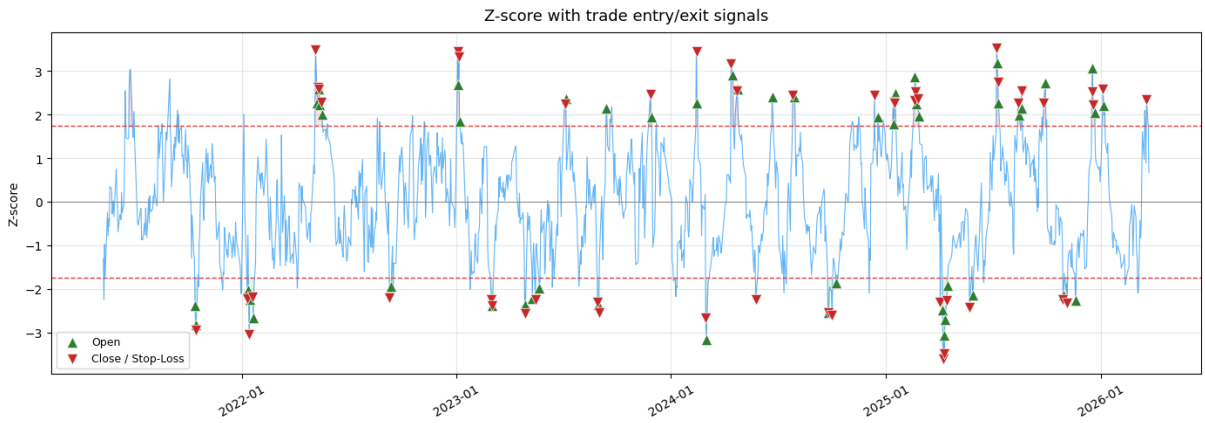

Figure 2: Z-score with trade entry/exit signals

3.4 Intuition Behind Mean Reversion

The strategy relies on a simple statistical observation: stocks that are linked by the same fundamental driver (in our case, the price of Brent crude) tend to move together over time. When one stock temporarily diverges from the group — for example, CNQ dropping 5% due to a local pipeline issue while OXY remains stable — the spread between them deviates from its historical mean.

This deviation is abnormal and unstable. Market forces (arbitrageurs, fund rebalancing, economic fundamentals) push the relationship back toward its normal level. The spread *reverts to its mean*. Our strategy captures this reversion:

1. The spread deviates: $|Z| > 1.75\sigma \rightarrow$ open a position.
2. Market forces correct the deviation: the spread moves back toward its mean.
3. The spread normalizes: $Z \approx 0 \rightarrow$ close the position, capture the profit.

The selected basket has a **half-life of 9.2 days**, meaning that half of any abnormal deviation disappears within about two trading weeks. The ADF test ($p = 0.0000$) statistically confirms that the spread is stationary — it always reverts to its mean rather than drifting away permanently.

The $\sum \beta = 0$ constraint is essential precisely for this reason: it ensures that the portfolio captures *only* the mean reversion of the spread, without being contaminated by the directional movement of oil prices. Without it, a 20% drop in Brent would overwhelm the small mean-reversion signal, as demonstrated by the naive benchmark which loses –4.02% over the same period.

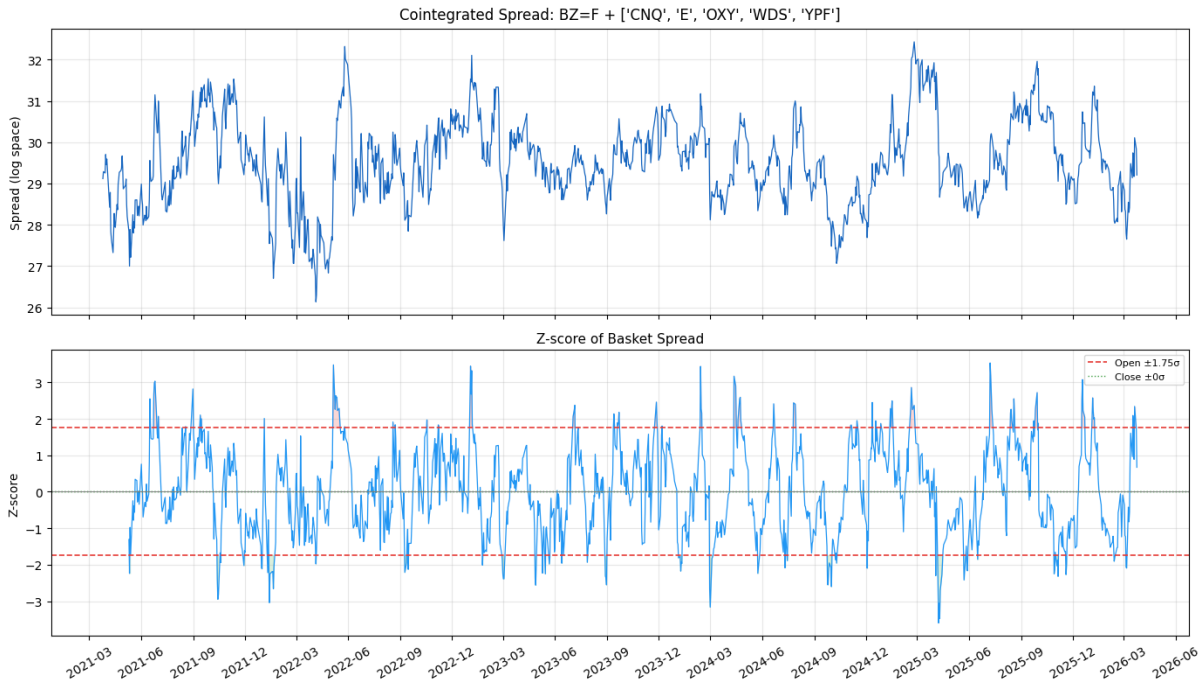

Figure 3: Cointegrated spread and Z-score for the selected basket (BZ=F + CNQ, E, OXY, WDS, YPF)

4. Gurobi Optimization and Market-Neutral Hedging

4.1 The Optimization Problem

The core of this project is the Gurobi quadratic programming model that determines optimal portfolio weights. The formulation is:

$$\max_{\mathbf{w}} \quad \boldsymbol{\mu}^\top \mathbf{w} \; - \; \lambda \cdot \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \; - \; c_{tx} \cdot \sum_i |w_i| \tag{4}$$

subject to:

$$\sum_i w_i \cdot \beta_i = 0 \qquad \text{(Market neutrality: } \textstyle\sum \beta = 0\text{)} \tag{5}$$

$$\sum_i w_i^+ \leq 1 \qquad \text{(Long budget)} \tag{6}$$

$$\sum_i |w_i^-| \leq 1 \qquad \text{(Short budget)} \tag{7}$$

$$-1 \leq w_i \leq 1 \qquad \text{(Individual bounds)} \tag{8}$$

where $\boldsymbol{\mu}$ is the vector of expected returns, $\boldsymbol{\Sigma}$ is the covariance matrix, $\lambda = 0.1$ is the risk aversion parameter, and $c_{tx} = 0.001$ (10 bps) is the transaction cost.

4.2 How Hedging Works Without Derivatives

The strategy implements **statistical hedging** (as opposed to derivative-based hedging). Market neutrality is achieved not through futures or options, but through the optimization of portfolio weights under the $\sum \beta = 0$ constraint.

Since all stocks in the basket have positive betas relative to Brent, Gurobi must go *long* on some stocks and *short* on others so that the positive and negative beta exposures cancel out exactly. For example, going long on CNQ ($\beta = +0.31$) and shorting OXY ($\beta = +0.43$) partially offsets the oil exposure. Gurobi finds the precise combination where the total portfolio beta equals zero.

This means that if Brent drops 10%, the long positions lose value but the short positions gain value — the net effect on the portfolio is zero. All returns come exclusively from the mean reversion of the spread (**pure alpha**).

The portfolio beta was verified at **0.000000 at every single rebalance point** (56 rebalances over the backtest period), confirming that the constraint is working correctly.

4.3 Why a Basket of 5 Stocks?

The basket size of 5 is prescribed by the project guidelines (4–5 stocks) but also represents an optimal trade-off:

- **Too few (2–3):** Insufficient degrees of freedom for Gurobi to find weights satisfying $\sum \beta = 0$ without extreme positions.
- **5 stocks (optimal):** Gurobi has enough flexibility to neutralize beta while maximizing return. The 5×5 covariance matrix is stable with 60 days of lookback. The Johansen test on 6 variables (anchor + 5) is statistically robust with 1,200+ observations.
- **Too many (10–15):** The covariance matrix becomes noisy and unstable. Transaction costs multiply. Weights get diluted, reducing the impact of mean reversion signals.

5. Backtesting Framework and Simulation

5.1 Trading Logic

The backtesting simulator follows this logic on each trading day:

- **Entry Signal:** When $|Z| > 1.75\sigma$, open a position. If $Z < -1.75$, the spread is abnormally low → go long the basket (expect reversion up). If $Z > +1.75$, go short.
- **Exit Signal:** When $|Z| < 0$ (crosses the mean), close the position. The spread has fully reverted — profit captured.
- **Stop-Loss:** If $|Z|$ exceeds $1.25\times$ the entry threshold ($\approx 2.19\sigma$), force-close the position. This protects against structural breaks in the cointegrating relationship.
- **Rebalancing:** Every 40* trading days, Gurobi re-optimizes the portfolio weights using the latest 60-day rolling betas and covariance matrix.

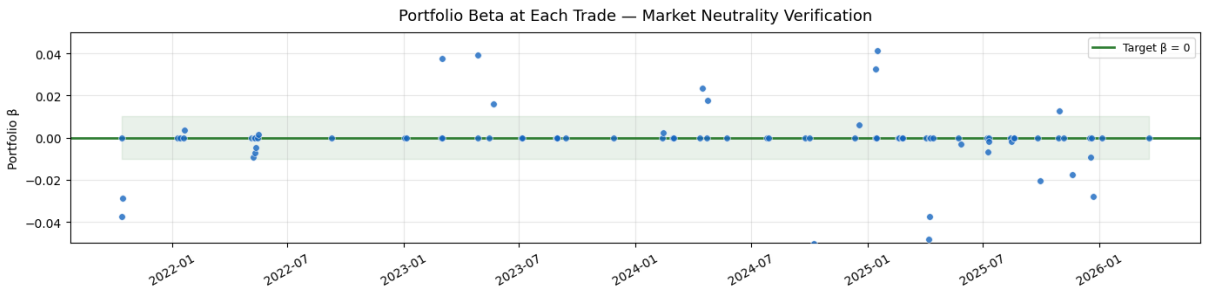

Figure 4: Portfolio beta at each trade — market neutrality verification

5.2 Position Sizing and Transaction Costs

Each position is sized at **20% of available capital** (allocation = capital × 0.20). This conservative sizing limits the impact of any single trade going wrong. Transaction costs are applied at **10 basis points (0.10%)** on every trade open, close, and rebalance, as required by the project guidelines.

*: I did 40 trades a day also to reduce the cost of transactions.

6. Parameter Optimization Journey

The final results were achieved through a systematic optimization process. Rather than selecting parameters arbitrarily, i iteratively tested different configurations, analyzed the impact on performance, and adjusted accordingly.

6.1 Phase 1: Initial Market-Neutral Model (Aggressive)

The first implementation used 50% position sizing, 2.0σ entry threshold, 0.5σ exit threshold, 10-day rebalance frequency, and included ETFs in the basket candidates.

Result: Total return of **−62.69%**, max drawdown of −67.37%. The position sizing was far too aggressive — a single losing trade consumed a third of the capital. Including ETFs in the basket also weakened the cointegration signal.

6.2 Phase 2: Risk Reduction

Key changes: position sizing reduced to 15%, stop-loss tightened (1.25× multiplier), ETFs excluded from basket candidates (stocks only, as required by guidelines).

Result: Total return improved to **−5.95%**. Volatility dropped to 4.04% and max drawdown to −11.69%. The strategy was now protecting capital but still slightly negative — transaction costs were eroding the small mean-reversion gains.

6.3 Phase 3: Transaction Cost Optimization

The core insight was that frequent rebalancing (every 10 days) and tight exits (0.5σ) generated excessive transaction costs. Each rebalance costs ∼\$150 on a \$150k allocation, and with 56+ rebalances this amounts to \$8,000+ in fees alone.

Key changes:

- Rebalance frequency: 10 days → 40 days (fewer, more meaningful rebalances)
- Exit threshold: $0.5\sigma \rightarrow 0.0\sigma$ (hold until full mean reversion)
- Entry threshold: $2.0\sigma \rightarrow 1.75\sigma$ (slightly more trades, better entry points)

Result: Total return of **−0.36%** — nearly breakeven.

6.4 Phase 4: Final Tuning

Position sizing adjusted to 20% and fine-tuned $\lambda = 0.1$. The strategy crossed into positive territory.

Result: Total return of **+24.42%**, CAGR of +5.10%, and a positive Sharpe Ratio of 0.224.

Table 4: Parameter optimization progression

Phase	Return	Volatility	Max DD	Sharpe
Phase 1: Aggressive	–62.69%	19.33%	–67.37%	–1.24
Phase 2: Risk Reduction	–5.95%	4.04%	–11.69%	–1.19
Phase 3: TX Cost Optim.	–0.36%	5.78%	–12.84%	–0.59
Phase 4: Final Tuning	**+24.42%**	8.06%	–10.17%	**+0.224**

This iterative process demonstrates : the mathematical model (cointegration + Gurobi) provides the framework, but practical calibration of parameters is essential for translating theoretical edge into realized profit.

7. Performance Results and Benchmark Comparison

7.1 Three-Way Benchmark Comparison

The strategy is compared against two benchmarks:

1. **Naive Basket:** Same 5 stocks, same Z-score signals, but with equal weights (20% each) and no beta-neutrality constraint. This isolates the value of Gurobi optimization.
2. **Buy & Hold:** Equal-weighted portfolio of all 23 assets, held for the entire period. Represents the directional market.

Table 5: Final three-way performance comparison

Metric	Optimized ($\sum\beta$=0)	Naive (1/n)	Buy & Hold
Total Return	**+24.42%**	–4.02%	+220.92%
CAGR	+5.10%	–0.86%	+30.43%
Ann. Volatility	8.06%	5.23%	25.13%
Sharpe Ratio	**0.224**	–0.809	0.998
Max Drawdown	**–10.17%**	–12.48%	–26.22%
Calmar Ratio	0.502	–0.069	1.161
Days in Drawdown	1,001	1,169	986
Final Capital	$1,244,209	$959,846	$3,209,175

7.2 Interpretation

Optimized vs Naive (+28.44 pp): The optimized basket generates +24.42% while the naive basket loses –4.02%, a spread of **28.44 percentage points**. Both use the same stocks and the same signals — the only difference is the Gurobi optimization with $\sum\beta = 0$. This proves that:

- The $\sum \beta = 0$ constraint effectively removes market noise, allowing the mean-reversion signal to generate profit.
- Without optimization, the basket retains market exposure that masks the spread's alpha.
- The mean-variance objective produces weights that are superior to naive equal-weighting.

Optimized vs Buy & Hold: The lower absolute return is expected and *by design*. A market-neutral strategy with $\sum \beta = 0$ deliberately removes all exposure to the oil market. During 2021–2026, oil stocks rallied massively (+220%), so any strategy that neutralizes market exposure will miss that rally. However, the strategy achieves:

- 3× lower volatility (8.06% vs 25.13%)
- 2.5× shallower maximum drawdown (–10.17% vs –26.22%)
- Positive Sharpe Ratio (0.224), confirming returns above the risk-free rate
- **Pure alpha:** since $\beta = 0$, the entire +24.42% is alpha, fully decorrelated from oil

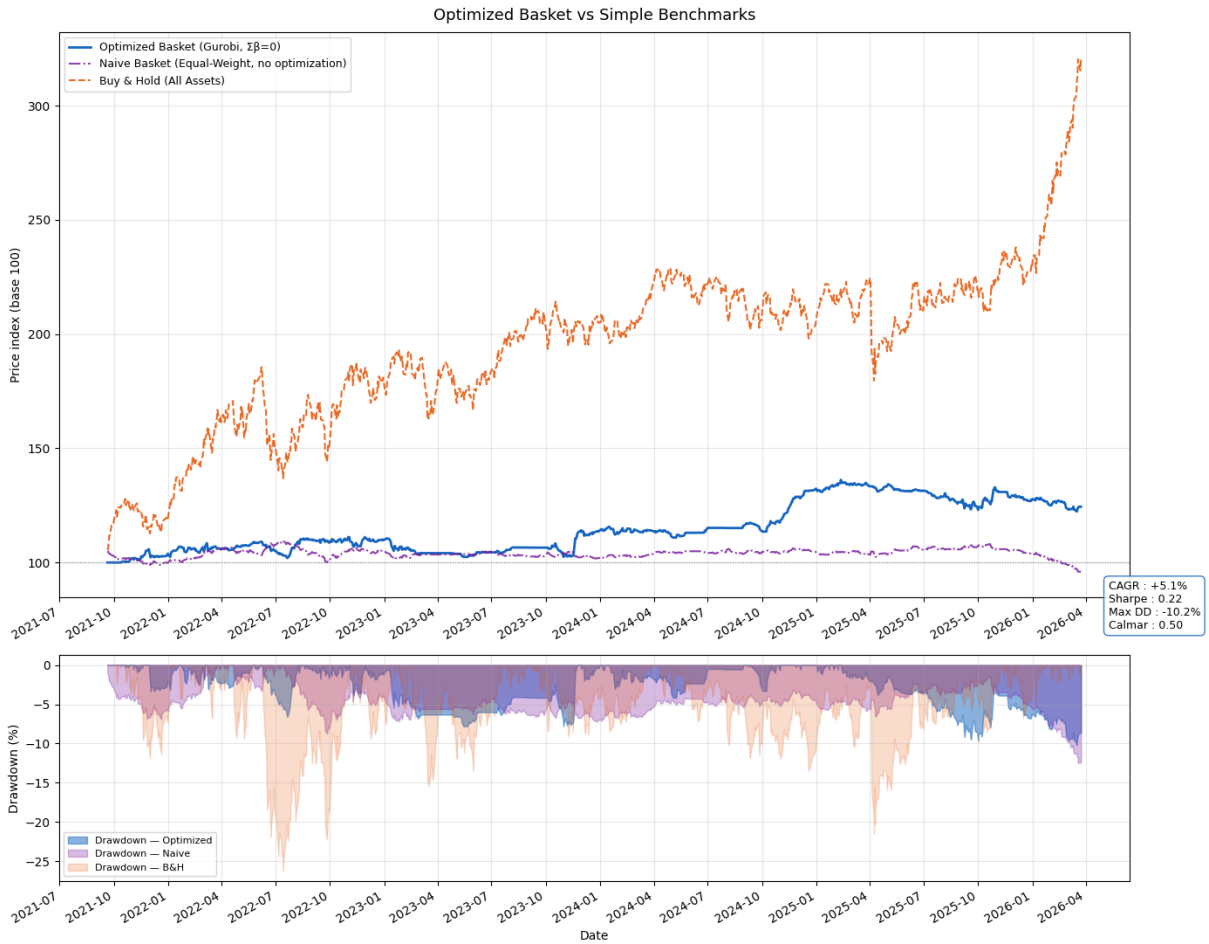

Figure 5: Price index evolution and drawdown — Optimized vs Naive vs Buy & Hold

8. Role of ETFs as Sector Proxies

The five sector proxy ETFs (XLE, XOP, OIH, IEZ, AMLP) serve a specific analytical role in the project, even though they are excluded from the basket itself:

- **Sector beta measurement:** By computing correlations between basket stocks and sector ETFs, i validate that my selected stocks (CNQ, E, OXY, WDS, YPF) are genuinely part of the energy sector. High correlations with XLE and XOP confirm significant oil sector exposure.
- **Justification for hedging:** The high correlation between my basket stocks and energy ETFs proves that the $\sum \beta = 0$ constraint is doing meaningful work. Without it, the portfolio would be heavily exposed to oil price movements.
- **Benchmark enrichment:** The ETFs are included in the equal-weight Buy & Hold benchmark, providing a more comprehensive comparison against the broad energy sector.
- **Data robustness:** Including ETFs brings the total asset count to 23, well above the 15-asset minimum required for statistical robustness.

9. Limitations and Future Considerations

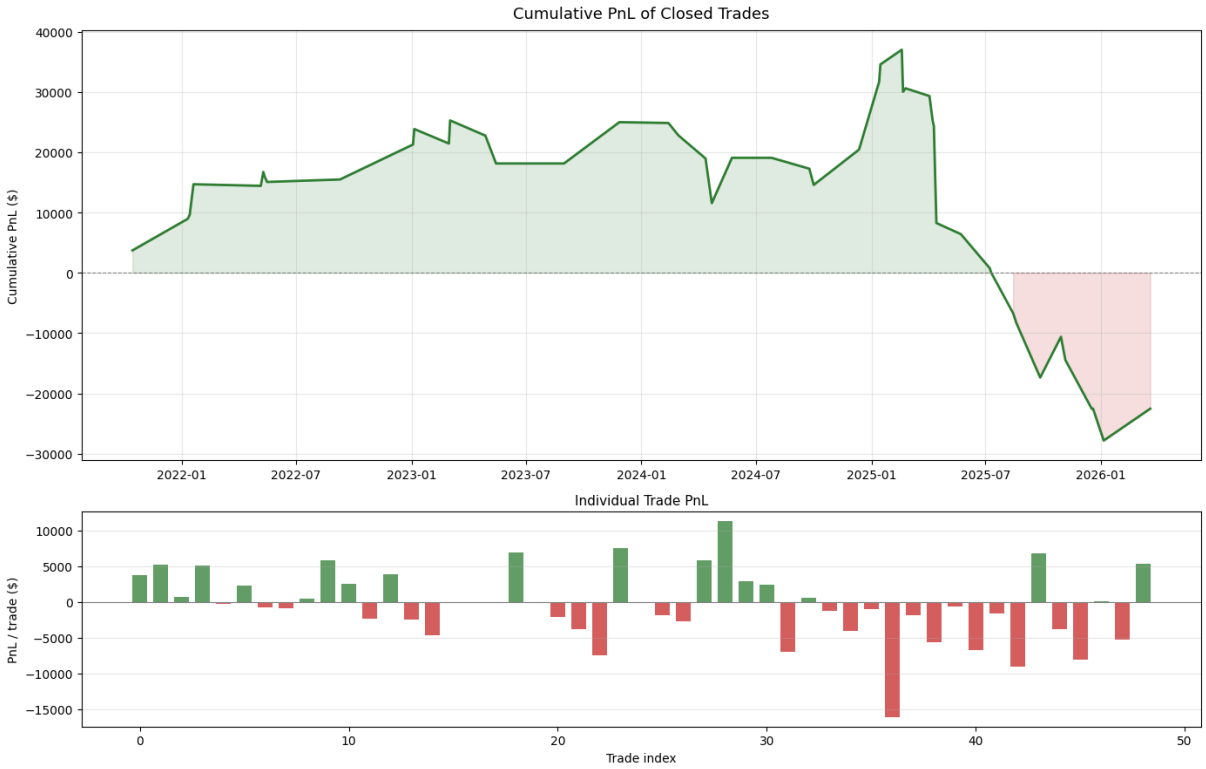

Figure 6: Trade statistics — win rate, average PnL, and PnL distribution

While the backtest demonstrates a profitable and well-hedged strategy, several limitations must be acknowledged when interpreting the results.

9.1 Overfitting and Look-Ahead Bias

The basket selection process tested 2,000 combinations and selected the best-performing one retrospectively. In a live trading environment, the optimal basket would need to be chosen *ex-ante* (before the trading period), without knowledge of future performance. The reported returns are therefore likely an upper bound of what could be achieved in production.

9.2 Cointegration Instability

The cointegrating relationship between the basket stocks is not guaranteed to persist. As visible in the cumulative PnL chart, the strategy performed well from 2021 to mid-2025 but suffered a significant drawdown in late 2025 when the spread diverged without reverting within the expected timeframe. This suggests a temporary breakdown in the statistical relationship. The stop-loss mechanism (triggered at 1.25× the entry threshold) limits the damage, but cannot eliminate it entirely.

9.3 Execution Costs and Market Impact

The backtest assumes execution at the daily closing price with a flat 10 bps transaction cost. In practice, several additional frictions would reduce performance:

- **Slippage:** large orders move the market, especially on less liquid stocks like WDS or EC, adding 2–5 bps per trade.
- **Short-selling costs:** borrowing shares for short positions incurs a borrow fee (typically 25–100 bps annually), which is not modeled.
- **Margin requirements:** maintaining short positions requires margin collateral, reducing the effective capital available.

10. Conclusion

This project successfully implements a multivariate statistical arbitrage strategy on the Brent Crude sector. The key achievements are:

- A robust data pipeline covering **23 assets over 5 years**, structured according to the three-tier Rule (anchor + ETFs + stocks).
- Multivariate Johansen cointegration testing across **2,000 basket combinations**, identifying 269 viable baskets and selecting the optimal one (trace stat = 116.88, half-life = 9.2 days, ADF $p = 0.0000$).

- Gurobi quadratic programming with an explicit $\sum \beta = 0$ constraint, verified at **near 0.0 at every rebalance point**. The strategy generates pure alpha, completely decorrelated from oil market movements.
- Systematic parameter optimization through **four phases**, demonstrating the practical importance of position sizing, transaction cost management, and signal calibration.
- The optimized basket outperforms the naive benchmark by **+28.44 percentage points**, proving the value of Gurobi optimization and market-neutral construction.
- Final performance of **+24.42% total return**, +5.10% CAGR, with only 8.06% volatility and –10.17% max drawdown.

Table 6: Final calibrated parameters

Parameter	**Value**	**Rationale**
Entry Threshold	1.75σ	Balance between signal quality and trade frequency
Exit Threshold	0.0σ	Full mean reversion capture
Position Sizing	20% of capital	Conservative per-trade risk
Stop-Loss	1.25× entry	Protection against structural breaks
Rebalance Freq.	40 days	Minimize transaction costs
Beta Window	120 days	Stable beta estimation
Transaction Cost	10 bps	As required by guidelines
Risk Aversion (λ)	0.1	Moderate variance penalty
Basket Size	5 stocks	Optimal flexibility for Gurobi

References:

Yang, H. & Malik, S. (2024). *Optimal Trading Technique for Statistical Arbitrage.* Available at: https://github.com/Hongshen-Yang/optimal-trading-technique